jaskier_door_lock variant C: two 12 V linear-actuator door locks on a socketed ESP32-C3 SuperMini,
with an optional RS485 (Modbus RTU) port. Docs: hardware/door-lock-c/README.md

ESP32-C3 SuperMini socket (antenna at the board edge):
IO4/IO3 door A IN1/IN2, IO5/IO6 door B IN1/IN2, IO1 door A contact, IO0 door B contact,
IO8/IO9 I2C SDA/SCL (pulled up: both strapping pins stay high at boot),
IO2 status LED (strapping pin, 10k pull-up), IO10 RS485 DE/RE (10k pull-down),
IO20 RS485 RO (UART0 RX), IO21 RS485 DI (UART0 TX), IO7 spare on J7.

RS485, optional: Modbus RTU slave for control without Wi-Fi.
DE/RE low at boot (R32), so the ROM boot log on IO21 never reaches the bus.
JP1 120 ohm termination: fit a jumper cap on the last device of the bus. J4 has two A/B/GND sets for a pass-through.
JP2/JP3 680 ohm fail-safe bias (bridge only if nothing else on the bus biases it).
D3 SM712 clamps A and B (-7 V / +12 V).

Door contact input: reed or micro switch to GND (closed = door shut). 1k + 100n filter, 10k pull-up.
Unconnected input reads high (open door). Swap M+/M- at the terminal if the bolt moves the wrong way.

Supply: 4S LiFePO4 house bank, 10-14.6 V. F1 1.1 A PTC, Q1 P-MOSFET reverse-polarity switch
(R1/R2 hold Vgs at -Vin/2, inside the AO3401A's 12 V limit), D1 16 V TVS: the INA219 bus input is 26 V max.
PS1 5 V module (SIP-3, 7805 pinout), D2 feeds the SuperMini's 5V pin, so its USB-C and the 12 V supply
never push into each other. The SuperMini makes the 3.3 V.

Each door: VBAT -> 0.1 R shunt -> DRV8871 VM. The INA219 reads motor current and the bank voltage.
Door A INA219 0x40 (A1 = A0 = GND), door B 0x44 (A1 = VS). Full on/off drive only, never PWM.
R_ILIM 56k: DRV8871 current limit 64 kV / 56k = 1.14 A, above the 0.55 A stall measured in the van.
4.7k pull-downs keep both bridge inputs low (coast) while the ESP32 resets or is unplugged.

DRV8871 + INA219 per door, Modbus RTU slave on RS485

Jaskier

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Title: Door lock C: two doors, ESP32-C3, RS485

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